

Chapter 7 — Evidence, Trust, Verification, Assurance, and Certification

Purpose and scope

Defines the evidence admission boundary, claims, provenance, custody, trust assessment, verification, assurance cases, and certification.

Source authority

Book I: Book I Articles II and XI; Decisions 22, 26, 34, 35, 40, 42, 43, 50, and 56

Book II: Book II Chapters 17, 18, 25, and 35

Book III: Book III Chapters 04, 06, 08, and 09

Normative semantic rules

- An Evidence Object **MUST** be immutable after admission; correction **MUST** occur through linked superseding, challenging, or explanatory objects.
- An Evidence Candidate or Audit Record **MUST NOT** be represented as an Evidence Object before governed admission.
- Verification **MUST** identify claims, criteria, methods, environment, evidence, uncertainty, and reproducibility.
- Certification **MUST** be scoped, time-bounded, attributable, evidence-based, suspendable, and revocable.

Canonical terms

HAL-TERM-0078 — Evidence Candidate

An observation, telemetry item, document, Audit Record, claim, or other integrity-protected input proposed for admission as an Evidence Object.

Distinction: It is not authoritative Evidence until accepted through the Evidence Service's governed admission and custody process.

Type/category/status: Pre-admission record / Evidence / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 26

Book II: Book II Chapters 18 and 25

Book III: Book III Chapters 4, 5, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0078` when it uses ****Evidence Candidate**** with this exact governed meaning: An observation, telemetry item, document, Audit Record, claim, or other integrity-protected input proposed for admission as an Evidence Object.

Counterexample: A dependent artifact uses ****Evidence Candidate**** in a way that violates its required distinction: It is not authoritative Evidence until accepted through the Evidence Service's governed admission and custody process.

Relationships: HAL-REL-0026

Lifecycle: HAL-TRANS-0014

Constraint: It is not authoritative Evidence until accepted through the Evidence Service's governed admission and custody process.

HAL-TERM-0079 — Evidence Object

An immutable object admitted and governed by the authoritative Evidence Service that records provenance, custody, source identity, observation or claim content, time, signatures, verification state, confidence, domain, and expiration metadata.

Distinction: An Evidence Object must not be mutated to revise a conclusion; later objects supersede or challenge it.

Type/category/status: Immutable provenance-bearing record / Evidence / Approved

Aliases: Evidence

Source basis: Direct source normalization

Book I: Book I Decision 26

Book II: Book II Chapters 18 and 25

Book III: Book III Chapters 4, 5, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0079` when it uses ****Evidence Object**** with this exact governed

meaning: An immutable object admitted and governed by the authoritative Evidence Service that records provenance, custody, source identity, observation or claim content, time, signatures, verification state, confidence, domain, and expiration metadata.

Counterexample: A dependent artifact uses **Evidence Object** in a way that violates its required distinction: An Evidence Object must not be mutated to revise a conclusion; later objects supersede or challenge it.

Relationships: HAL-REL-0026, HAL-REL-0027, HAL-REL-0028, HAL-REL-0053

Lifecycle: None registered

Constraint: An Evidence Object must not be mutated to revise a conclusion; later objects supersede or challenge it.

HAL-TERM-0080 — Audit Record

An append-only record of a protected action, authorization, decision, access, or change owned by the applicable audit domain.

Distinction: An Audit Record may produce an Evidence Candidate but is not automatically a general Evidence Object.

Type/category/status: Protected accountability record / Evidence / Approved

Aliases: audit log entry

Source basis: Direct source normalization

Book I: Book I Decision 26

Book II: Book II Chapters 18 and 25

Book III: Book III Chapters 4, 5, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0080` when it uses **Audit Record** with this exact governed meaning: An append-only record of a protected action, authorization, decision, access, or change owned by the applicable audit domain.

Counterexample: A dependent artifact uses **Audit Record** in a way that violates its required distinction: An Audit Record may produce an Evidence Candidate but is not automatically a general Evidence Object.

Relationships: None registered

Lifecycle: None registered

Constraint: An Audit Record may produce an Evidence Candidate but is not automatically a general Evidence Object.

HAL-TERM-0081 — Provenance

The attributable origin, derivation, transformation, and custody history of an artifact, datum, claim, model, decision, or Evidence Object.

Distinction: A source label without derivation and custody context is incomplete provenance.

Type/category/status: Origin history / Evidence / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 26

Book II: Book II Chapters 18 and 25

Book III: Book III Chapters 4, 5, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0081` when it uses **Provenance** with this exact governed meaning: The attributable origin, derivation, transformation, and custody history of an artifact, datum, claim, model, decision, or Evidence Object.

Counterexample: A dependent artifact uses **Provenance** in a way that violates its required distinction: A source label without derivation and custody context is incomplete provenance.

Relationships: None registered

Lifecycle: None registered

Constraint: A source label without derivation and custody context is incomplete provenance.

HAL-TERM-0082 — Chain of Custody

The ordered, attributable record of possession, control, transfer, and integrity protection for evidence or sensitive artifacts.

Distinction: It does not establish truth; it establishes accountable handling.

Type/category/status: Integrity history / Evidence / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 26

Book II: Book II Chapters 18 and 25

Book III: Book III Chapters 4, 5, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0082` when it uses **Chain of Custody** with this exact governed meaning: The ordered, attributable record of possession, control, transfer, and integrity protection for evidence or sensitive artifacts.

Counterexample: A dependent artifact uses **Chain of Custody** in a way that violates its required distinction: It does not establish truth; it establishes accountable handling.

Relationships: None registered

Lifecycle: None registered

Constraint: It does not establish truth; it establishes accountable handling.

HAL-TERM-0083 — Evidence Graph

A graph connecting Claims, Evidence Objects, sources, derivations, supporting or opposing relations, confidence, and conclusions.

Distinction: Graph connectivity does not make all linked material equally authoritative or trustworthy.

Type/category/status: Evidence relationship model / Evidence / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 26

Book II: Book II Chapters 18 and 25

Book III: Book III Chapters 4, 5, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0083` when it uses **Evidence Graph** with this exact governed meaning: A graph connecting Claims, Evidence Objects, sources, derivations, supporting or opposing relations, confidence, and conclusions.

Counterexample: A dependent artifact uses **Evidence Graph** in a way that violates its required distinction: Graph connectivity does not make all linked material equally authoritative or trustworthy.

Relationships: HAL-REL-0028

Lifecycle: None registered

Constraint: Graph connectivity does not make all linked material equally authoritative or trustworthy.

HAL-TERM-0084 — Verification

A reproducible, risk-scaled process that evaluates a Claim, invariant, behavior, artifact, or Outcome against explicit criteria.

Distinction: Verification produces evidence; it is not the same as certification or operational approval.

Type/category/status: Evidence-producing evaluation / Verification / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 43, 50, and 56

Book II: Book II Chapters 17 and 35

Book III: Book III Chapters 6, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0084` when it uses **Verification** with this exact governed meaning: A reproducible, risk-scaled process that evaluates a Claim, invariant, behavior, artifact, or Outcome against explicit criteria.

Counterexample: A dependent artifact uses **Verification** in a way that violates its required distinction: Verification produces evidence; it is not the same as certification or operational approval.

Relationships: HAL-REL-0029

Lifecycle: None registered

Constraint: Verification produces evidence; it is not the same as certification or operational approval.

HAL-TERM-0085 — Verification Plan

A governed specification of claims, risks, methods, environments, data, success criteria, independence, and required evidence for Verification.

Distinction: A test list without mapped claims and criteria is not a complete Verification Plan.

Type/category/status: Planning record / Verification / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 43, 50, and 56

Book II: Book II Chapters 17 and 35

Book III: Book III Chapters 6, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0085` when it uses **Verification Plan** with this exact governed meaning: A governed specification of claims, risks, methods, environments, data, success criteria, independence, and required evidence for Verification.

Counterexample: A dependent artifact uses **Verification Plan** in a way that violates its required distinction: A test list without mapped claims and criteria is not a complete Verification Plan.

Relationships: None registered

Lifecycle: None registered

Constraint: A test list without mapped claims and criteria is not a complete Verification Plan.

HAL-TERM-0086 — Assurance Case

A structured, reviewable argument connecting scoped claims to reasoning and sufficient supporting evidence.

Distinction: An Assurance Case must expose assumptions, defeaters, uncertainty, and evidence gaps.

Type/category/status: Structured argument / Verification / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 43, 50, and 56

Book II: Book II Chapters 17 and 35

Book III: Book III Chapters 6, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0086` when it uses **Assurance Case** with this exact governed meaning: A structured, reviewable argument connecting scoped claims to reasoning and sufficient supporting evidence.

Counterexample: A dependent artifact uses **Assurance Case** in a way that violates its required distinction: An Assurance Case must expose assumptions, defeaters, uncertainty, and evidence gaps.

Relationships: HAL-REL-0030, HAL-REL-0031

Lifecycle: None registered

Constraint: An Assurance Case must expose assumptions, defeaters, uncertainty, and evidence gaps.

HAL-TERM-0087 — Certification

A scoped, time-bounded, evidence-based determination by an authorized certifier that specified conformance claims are satisfied.

Distinction: Certification is not permanent, universal, or self-issued by the artifact being certified.

Type/category/status: Governed assurance decision / Verification / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 43, 50, and 56

Book II: Book II Chapters 17 and 35

Book III: Book III Chapters 6, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0087` when it uses **Certification** with this exact governed meaning: A scoped, time-bounded, evidence-based determination by an authorized certifier that specified conformance claims are satisfied.

Counterexample: A dependent artifact uses **Certification** in a way that violates its required distinction: Certification is not permanent, universal, or self-issued by the artifact being certified.

Relationships: HAL-REL-0031

Lifecycle: HAL-TRANS-0015, HAL-TRANS-0016, HAL-TRANS-0017

Constraint: Certification is not permanent, universal, or self-issued by the artifact being certified.

HAL-TERM-0088 — Conformance

The evidenced condition of satisfying identified requirements from identified authoritative sources within a declared scope and version.

Distinction: Conformance is never implied merely by compatibility, successful execution, or absence of known defects.

Type/category/status: Evaluated relation / Verification / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 43, 50, and 56

Book II: Book II Chapters 17 and 35

Book III: Book III Chapters 6, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0088` when it uses **Conformance** with this exact governed meaning: The evidenced condition of satisfying identified requirements from identified authoritative sources within a declared scope and version.

Counterexample: A dependent artifact uses **Conformance** in a way that violates its required distinction: Conformance is never implied merely by compatibility, successful execution, or absence of known defects.

Relationships: None registered

Lifecycle: None registered

Constraint: Conformance is never implied merely by compatibility, successful execution, or absence of known defects.

HAL-TERM-0089 — Confidence

A bounded assessment of support for a Claim or prediction, expressed with method, scope, evidence basis, uncertainty, and time sensitivity.

Distinction: Confidence is not probability unless a defined model justifies that interpretation.

Type/category/status: Calibrated assessment / Trust / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 43, 50, and 56

Book II: Book II Chapters 17 and 35

Book III: Book III Chapters 6, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0089` when it uses **Confidence** with this exact governed meaning: A bounded assessment of support for a Claim or prediction, expressed with method, scope, evidence basis, uncertainty, and time sensitivity.

Counterexample: A dependent artifact uses **Confidence** in a way that violates its required distinction: Confidence is not probability unless a defined model justifies that interpretation.

Relationships: None registered

Lifecycle: None registered

Constraint: Confidence is not probability unless a defined model justifies that interpretation.

HAL-TERM-0165 — Evidence Service

The Book II authoritative service that admits and governs Evidence Objects and owns their custody, signatures, provenance bindings, and verification state.

Distinction: Observability, audit, and source systems may produce Evidence Candidates or records but cannot independently admit Evidence Objects or mutate evidentiary meaning.

Type/category/status: Architectural component class / Evidence / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 26

Book II: Book II Chapters 18 and 25

Book III: Book III Chapters 4, 5, 6, and 8

Example: A telemetry record enters as an Evidence Candidate; the Evidence Service validates provenance and custody before admitting a new immutable Evidence Object.

Counterexample: A logging or observability service labels its mutable record an Evidence Object without the governed admission process.

Relationships: HAL-REL-0053

Lifecycle: None registered

Constraint: Observability, audit, and source systems may produce Evidence Candidates or records but cannot independently admit Evidence Objects or mutate evidentiary meaning.

Relationship and lifecycle rules

Relationships involving these terms **MUST** use the typed relationship register. Lifecycle-bearing concepts **MUST** use the governed transition register; a status label alone never authorizes a prohibited transition.

Term-specific semantic evidence

Every Term Record above contains a governed example, counterexample, constraint, relationship reference set, and lifecycle reference set. "None registered" is explicit.

Anti-patterns

Semantic drift, authority laundering, and collapsing an entity with its identifier, record, attributes, role, or state are prohibited.

Verification

Verify ID and label uniqueness, precise source traceability, source-basis classification, relationship and lifecycle consistency, ambiguity controls, machine-readable parity, and cross-format content parity.

Change and deprecation

Every semantic change requires authority, compatibility, dependency impact, migration, reviewer, effective version, and any replacement and sunset conditions.

Review findings

Constitutional, architecture, engineering, semantic, usability, parity, and Owner-threshold reviews passed. No Owner Review item.